CISS245: Advanced Programming
Quiz q2901

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Q1. Compute by hand the value of $f(4)$ if $f(n)$ satisfies the following:

$$f(n) = \begin{cases} 1 & \text{if } n = 0 \\ 2 & \text{if } n = 1 \\ 2f(n-2) - 3 & \text{if } n \geq 2 \end{cases}$$

Answer:

```
-5
```

Q2. Compute by hand the value of c if $f(4) = 10$ and $f(n)$ satisfies the following:

$$f(n) = \begin{cases} c & \text{if } n = 0 \\ 2 & \text{if } n = 1 \\ f(n-1) + 2f(n-2) + 1 & \text{if } n \geq 2 \end{cases}$$

Your answer below will be considered a mathematical expression, not a C++ expression. So when you write 1/3 below, you mean "one third" and not 0.
Answer:

```
-5/6
```

Q3. Write a C++ function that implements the following

$$f(n) = \begin{cases} 1 & \text{if } n = 0 \\ 2 & \text{if } n = 1 \\ 3f(n-1) + 4f(n-2) & \text{if } n \geq 2 \end{cases}$$

Answer:

```
int f(int n)
{
  if (n == 0) return 1;
  else if (n == 1) return 2;
  else return 3 * f(n - 1) + 4 * f(n - 2);
}
```

Q4. Write a C++ function that implements the computation of x^n where x is a `double` and `n` is an integer ≥ 0. The following should help with your recursion:

$$x^6 = x^5 \cdot x$$

(The idea is similar to the recursive factorial computation.)
Answer:

```
double power(double x, int n)
{
  if (n == 0) return 1.0;
  else return power(x, n - 1) * x;
}
```

Q5. Write a C++ function that implements the recursive GCD. (The one in the notes.)
Answer:

```
int gcd(int a, int b)
{
  if (b == 0) return a;
  else return gcd(b, a % b);
}
```

Q6. Write a recursive C++ function that computes

$$f(n) = \frac{0}{2^0} + \frac{1}{2^1} + \frac{2}{2^2} + \frac{3}{2^3} + \cdots + \frac{n}{2^n}$$

(You want to think about two things: What is $f(0)$? What is the relationship between $f(n+1)$ in terms of $f(n)$?)
Answer:

```
double f(int n)
{
  if (n == 0) return 0.0;
  else return f(n - 1) + n / pow(2.0, n);
}
```

Instructions

In the file `thispreamble.tex` look for

```
\renewcommand\AUTHOR{}
```

and enter your email address:

```
\renewcommand\AUTHOR{jdoe5@cougars.ccis.edu}
```

(This is not really necessary since alex will change that for you when you execute `make`.) In your bash shell, execute "`make`" to recompile `main.pdf`. Execute "`make v`" to view `main.pdf`.

Enter your answers in `main.tex`. In the bash shell, execute "`make`" to recompile `main.pdf`. Execute "`make v`" to view `main.pdf`.

For each question, you'll see boxes for you to fill. For small boxes, if you see

```
1 + 1 = \answerbox{}.
```

you do this:

```
1 + 1 = \answerbox{2}.
```

`answerbox` will also appear in "true/false" and "multiple-choice" questions.

For longer answers that need typewriter font, if you see

```
Write a C++ statement that declares an integer variable name x.
\begin{answercode}
\end{answercode}
```

you do this:

```
Write a C++ statement that declares an integer variable name x.
\begin{answercode}
int x;
\end{answercode}
```

`answercode` will appear in questions asking for code, algorithm, and program output. In this case, indentation and spacing is significant. For program output, I do look at spaces and newlines.

For long answers (not in typewriter font) if you see

```
What is the color of the sky?
\begin{answerlong}
\end{answerlong}
```

you can write

```
What is the color of the sky?
\begin{answerlong}
The color of the sky is blue.
\end{answerlong}
```

A question that begins with "T or F or M" requires you to identify whether it is true or false, or meaningless. "Meaningless" means something's wrong with the question and it is not well-defined. Something like "$1 + 2 = 4$" is either true or false (of course it's false). Something like "$1+_2 = 4$?" does not make sense.

When writing results of computations, make sure it's simplified. For instance write 2 instead of $1 + 1$.

Higher level classes.

For students beyond 245: You can put LaTeX commands in `answerlong`.

More examples of meaningless statements: Questions such as "Is $42 = 1+_2$ true or false?" or "Is $42 = \{2\}^{\{3\}}$ true or false?" does not make sense. "Is $P(42) = \{42\}$ true or false?" is meaningless because $P(X)$ is only defined if X is a set. For "Is $1 + 2 + 3$ true or false?", "$1 + 2 + 3$" is well–defined but as a "numerical expression", not as a "proposition", i.e., it cannot be true or false. Therefore "Is $1 + 2 + 3$ true or false?" is also not a well-defined question.

More examples of simplification: When you write down sets, if the answer is $\{1\}$, do not write $\{1, 1\}$. And when the values can be ordered, write the elements of the set in ascending order. When writing polynomials, begin with the highest degree term.

When writing a counterexample, always write the simplest.